

BPMN Process Modeling Assignment

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Repository : <https://github.com/animeshjha01/BPMN-Process-Modeling-Assignment-1.git>

Branch : main

Tool : Camunda Modeler 5.49.0 / Camunda 8.9

This report documents the three BPMN models currently uploaded to the GitHub repository. The descriptions and element lists below are aligned with the actual BPMN files in the repository: diagram_1.bpmn, diagram_2.bpmn, and diagram_3.bpmn.

1. Repository Contents

File	Scenario
diagram_1.bpmn	Employee Leave Approval
diagram_2.bpmn	Online Purchase Order Processing
diagram_3.bpmn	IT Service Request

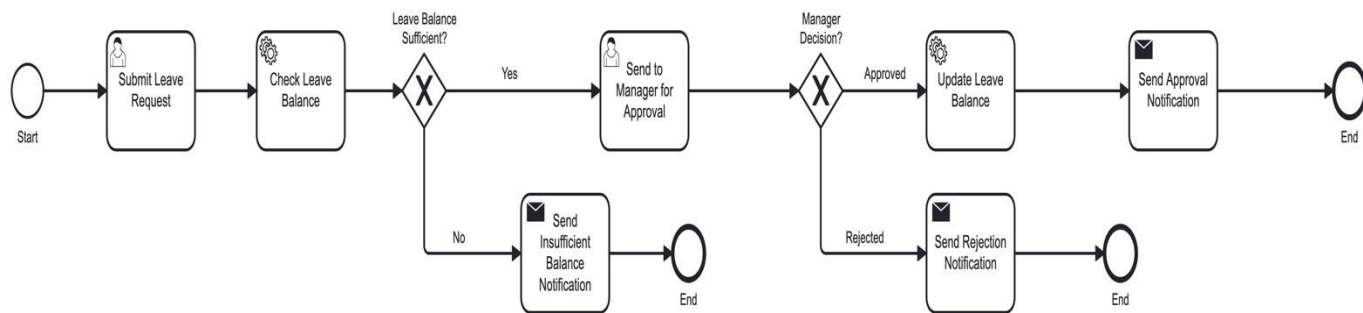
2. Scenario 1 Employee Leave Approval

The employee leave process starts with a leave request. The system checks the available leave balance. An Exclusive Gateway decides whether sufficient balance exists. If the balance is insufficient, a Send Task sends an insufficient-balance notification and the process ends. If sufficient, the request is sent to the manager. A second Exclusive Gateway handles the manager's approval or rejection. A approved request updates the leave balance and sends an approval notification; a rejected request sends a rejection notification.

Elements present in diagram_1.bpmn

BPMN Element	Actual element in the uploaded model
Start Event	Start
User Task	Submit Leave Request
Service Task	Check Leave Balance
Exclusive Gateway	Leave Balance Sufficient?
User Task	Send to Manager for Approval
Exclusive Gateway	Manager Decision?
Service Task	Update Leave Balance
Send Task	Send Approval Notification
Send Task	Send Rejection Notification
Send Task	Send Insufficient Balance Notification
End Events	Three End events for the three outcomes

Decision paths: Leave Balance Sufficient = Yes → Send to Manager for Approval. No → Send Insufficient Balance Notification → End. Manager Decision = Approved → Update Leave Balance → Send Approval Notification → End. Rejected → Send Rejection Notification → End.



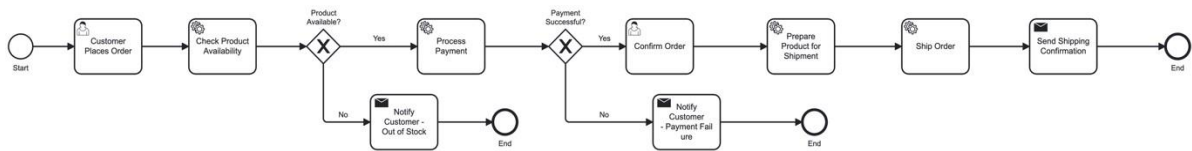
3. Scenario 2 Online Purchase Order Processing

The order process begins when the customer places an order. Product availability is checked and the first Exclusive Gateway selects the available or unavailable path. An unavailable product produces an out-of-stock notification and ends the process. When available, payment is processed. The second Exclusive Gateway checks payment success. A failed payment produces a payment-failure notification and ends the process. A successful payment leads to order confirmation, product preparation, shipping, and a shipping confirmation.

Elements present in diagram_2.bpmn

BPMN Element	Actual element in the uploaded model
Start Event	Start
User Task	Customer Places Order
Service Task	Check Product Availability
Exclusive Gateway	Product Available?
Service Task	Process Payment
Exclusive Gateway	Payment Successful?
User Task	Confirm Order
Service Task	Prepare Product for Shipment
Service Task	Ship Order
Send Task	Notify Customer - Out of Stock
Send Task	Notify Customer - Payment Failure
Send Task	Send Shipping Confirmation
End Events	Three End events for completed and failed paths

Decision paths: Product Available = Yes → Process Payment; No → Notify Customer - Out of Stock → End. Payment Successful = Yes → Confirm Order → Prepare Product for Shipment → Ship Order → Send Shipping Confirmation → End. No → Notify Customer - Payment Failure → End.



4. Scenario 3 IT Service Request

The IT support process starts when an employee submits an IT support request. The request is registered and the problem severity is checked. The first Exclusive Gateway routes high-severity problems to a senior technician and low-severity problems to a support technician. Both paths continue to problem investigation. A second Exclusive Gateway determines whether the problem can be resolved internally. If yes, the problem is fixed. If no, it is escalated to an external service provider. Both paths then update the request status, send a resolution notification, and reach the End Event.

Elements present in diagram_3.bpmn

BPMN Element	Actual element in the uploaded model
Start Event	Start
User Task	Submit IT Support Request
Service Task	Register Request
Service Task	Check Problem Severity
Exclusive Gateway	Problem Severity?
User Task	Assign to Senior Technician
User Task	Assign to Support Technician
Service Task	Investigate Problem
Exclusive Gateway	Can Be Resolved?
Service Task	Fix Problem
Service Task	Escalate to External Service Provider
Service Task	Update Request Status
Send Task	Send Resolution Notification
End Event	End

Decision paths: Problem Severity = High → Assign to Senior Technician; Low → Assign to Support Technician. Can Be Resolved = Yes → Fix Problem; No → Escalate to External Service Provider. Both paths merge at Update Request Status.

